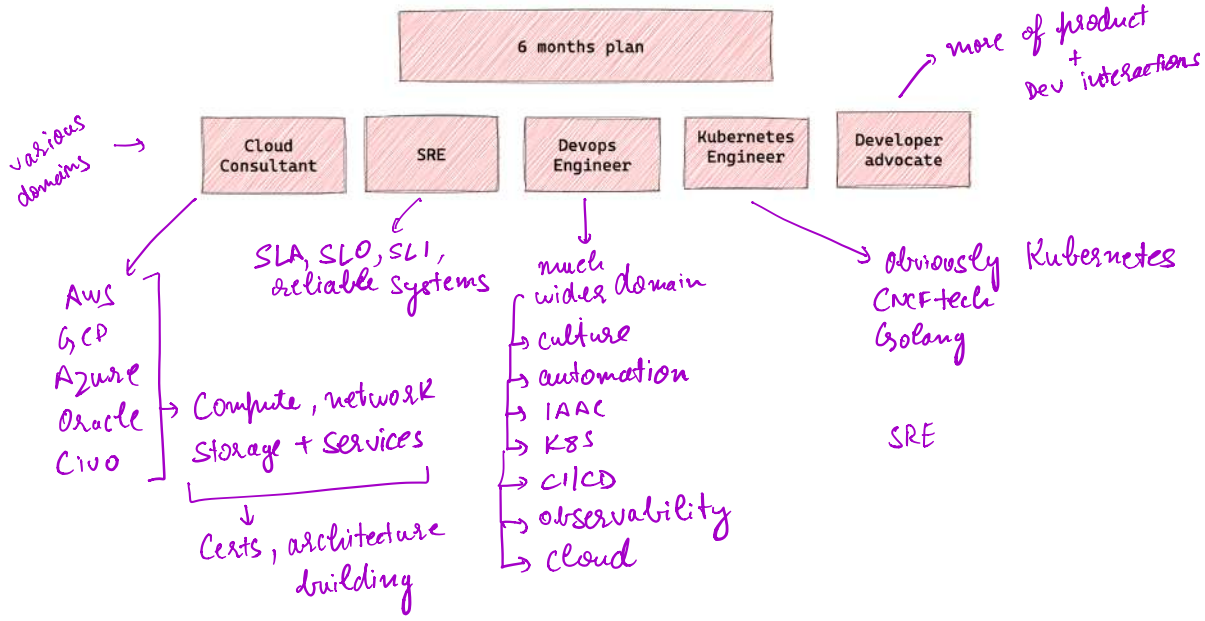
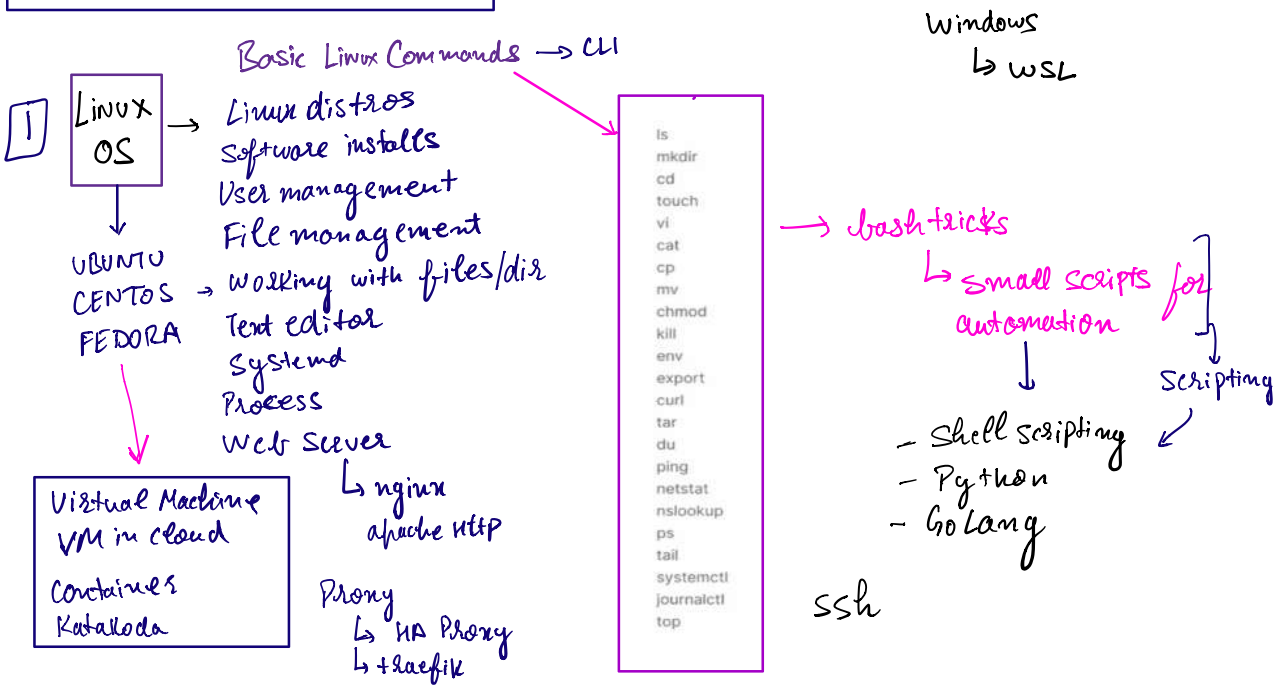
# Roadmap 2022

ANYONE CAN LEARN




# Prerequisites

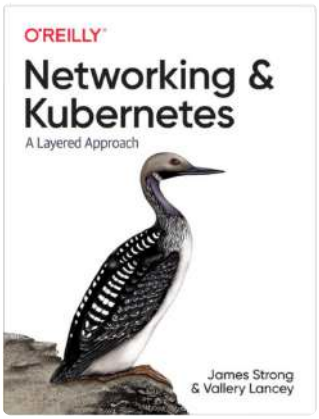


Resources → edx linux course by LF
Devops Pre requisites Course → Youtube

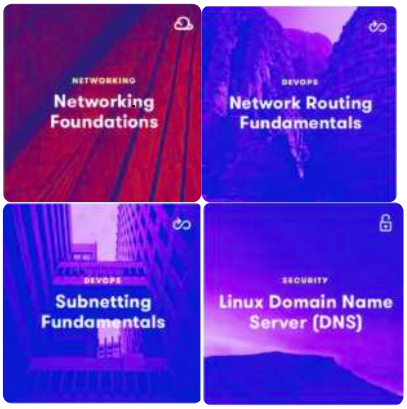


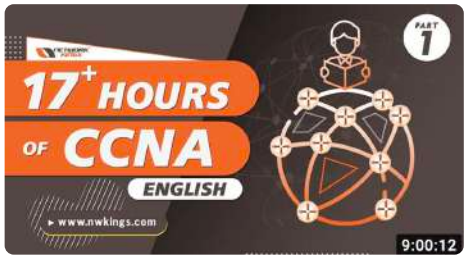

(2) Basic Networking

Resources



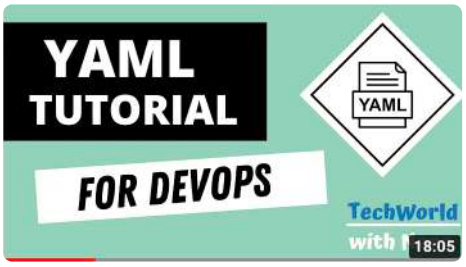

→ How system communicate
→ LAN / WAN
→ What is a switch, router, ISP ?
→ IP address and types
→ OSI model
→ Basic Subnetting
→ DNS basics
→ Switching & Routing




Part 1 (Enough)

(3) Introduction to YAML



(4) GIT → Source code
↳ has to live somewhere.

GitHub
GitLab

- How to work with git
  - setup
  - Create repo
  - commit and push code
  - Branching
  - PR
  - Public/Private repo
  - Pull, rebase, fork, clone

Resources
↳




Free codecamp
Youtube




Kunal Kushwaha
Youtube




Free codecamp
Youtube

5 CLOUD

↳ Any Cloud provider

Terms
→ IAAS
→ SAAS
→ PAAS

What is
- cloud Concepts
- Benefits
- Infra

Network
Storage
Compute
IAM
Billing

→ you have different services

→ You need cloud account
↓
use free credits for learning.

↗ Bigdata, Databases
ML, Kubernetes

GCP, Aws, Azure, Oracle

Civo → Cloud Native approach → managed Kubernetes

Resources



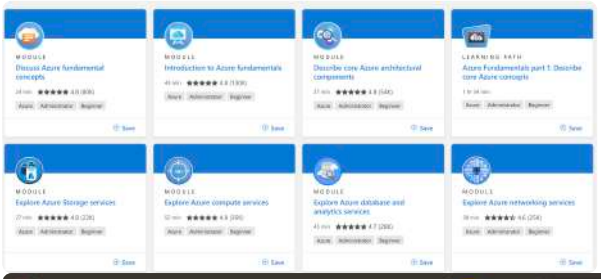

↓

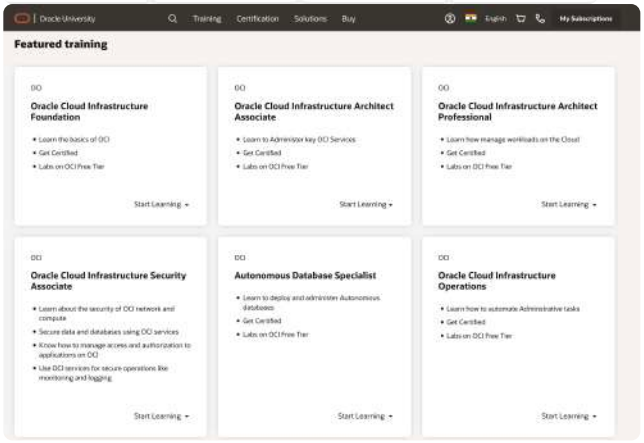
Associate

↓

Devops Professional

Azure ↴









GCP ↴

Oracle

→ You should be able to work with cloud

→ Learn quickly

→ Certs are addon

# (6) Virtualization and Containers

→ Servers → virtualization → containers

What is → VM → Container → Difference

Container technology →
- → Docker
- → Podman + Buildah
- → Kaniko

Build via buildah, run via podman

build from dockerfile inside container K8's cluster

→ Ko → go apps

→ kim → CRI, containerd, buildkit

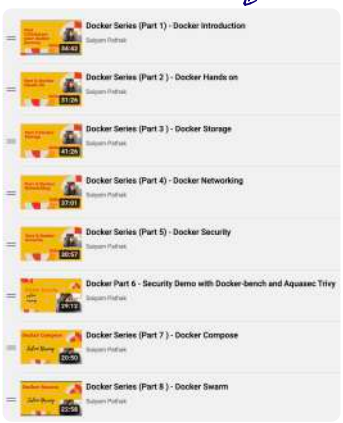
All these are open source projects.

How to
- build
- Push
- Pull
- tag
- Run

Concepts
- OCI
- CRI




Resources → Docker playlist (my youtube) ← Start simple → Docker (Many free courses)







Free Code Camp Youtube




Tech World with Nana




Dockercon




Docker deep dive (my youtube)

Advanced → Docker Security
- Dockerfile Best Practices
- Container scanning (Trivy)



Ivan Blog series







HackerSploit Youtube

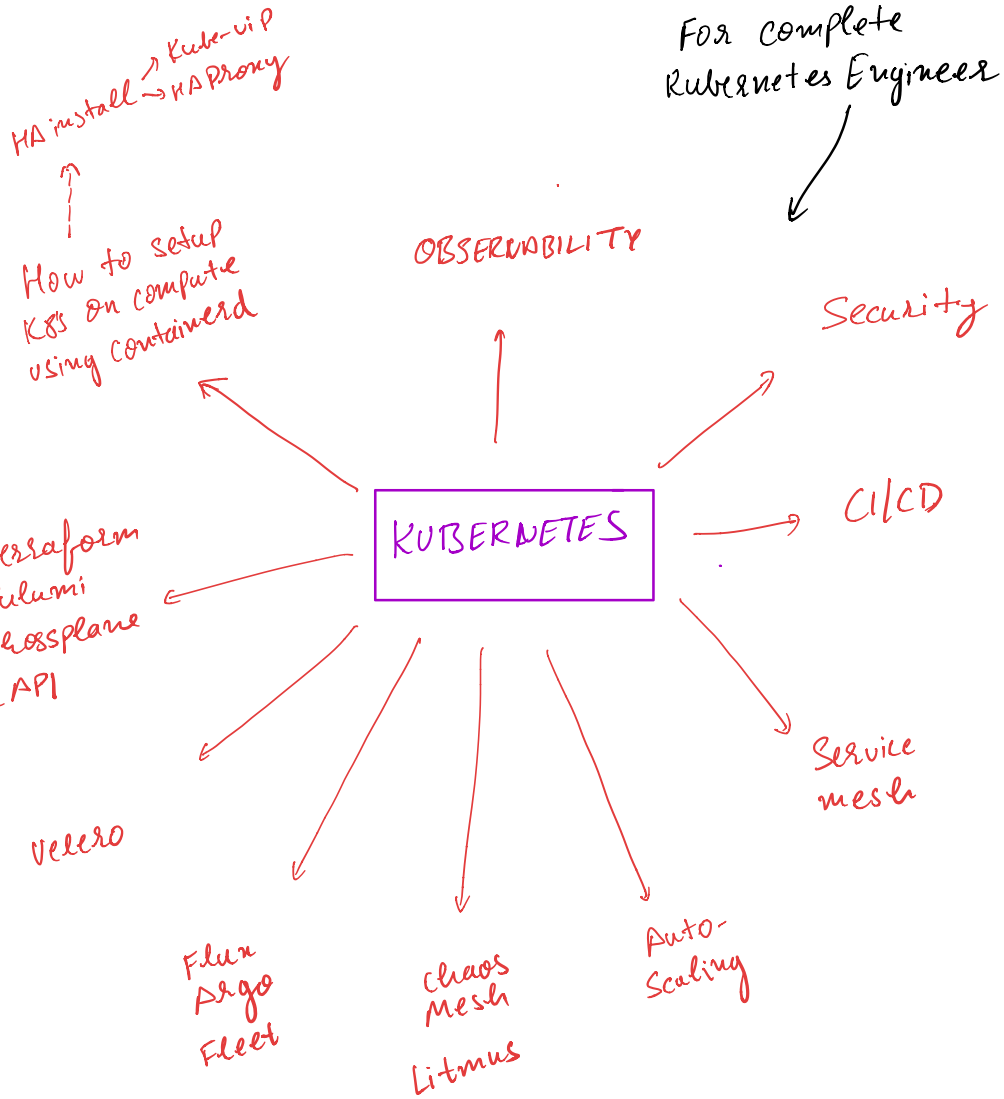
7 KUBERNETES → Container orchestration.

↓ ↘ a must learn for all domains

lot of Resources → What is

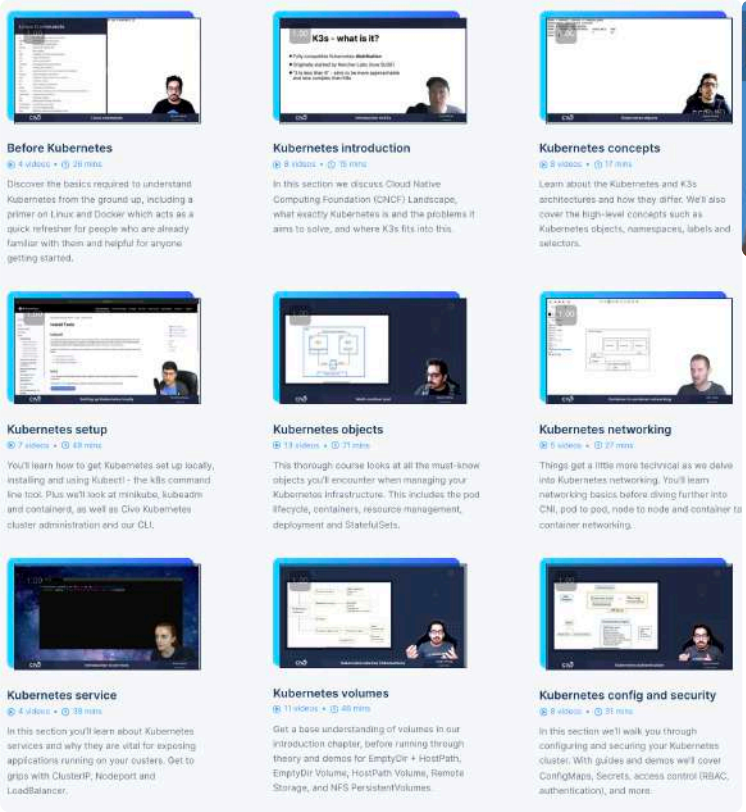
- K8's
- Architecture
- objects
- Secrets
- Configmaps
- PV, PVC
- Networking
- Services

Mind Map For complete Kubernetes Engineer

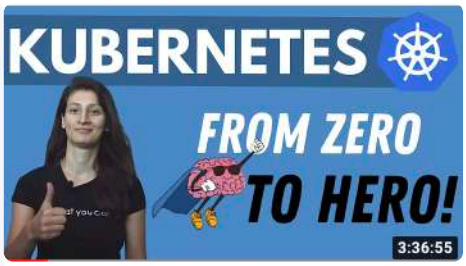


Resources

(1) → Civo Academy




(2)




Techworld with Nana
Youtube.

(4) Kube.academy →



(3) → edx, course by LF




Path to certification

→ KCNA, CKA, CKAD, CKS

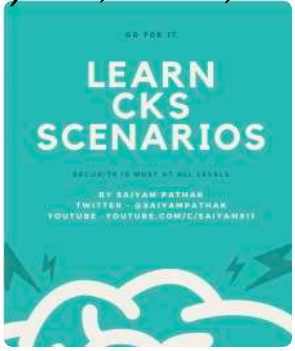



KCNA Kubernetes and cloud native associate certification



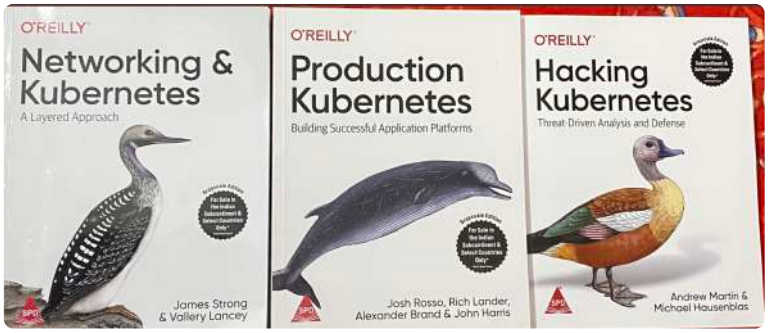

My Youtube (Many Resources)

→ Certs magic show
↓
CNCF official




My CKS Book

More depth Knowledge →




(8) CI/CD →

From Devops view

GitHub Actions
Jenkins

Code Commit → Build → Test → Stage → Deploy → Release

What are the Concepts → CI/CD Landscape → CDFoundation




Kubernetes Native → GitOps → Concepts + tooling → Flux CD, Argo CD, Fleet → My YouTube

More for Kubernetes Engineer + SRE

CI/CD → must watch from experts → Week



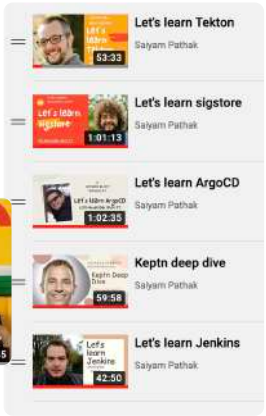

Simplilearn Youtube




Techworld with Nana




My Youtube



⑨ IAC → Infrastructure as Code

↳ IAC concepts

DO - Terraform

KE SRE - Crossplane

KE SRE - Pulumi

INFRA PROVISIONING

→ Code to create infra
→ automate → Creation, updating, deleting infra
→ no manual errors
→ no misconfiguration
→ state of your infra.




Terraform →




Crossplane → my Youtube
→ CNCFMinutes Crossplane
→ Crossplane deep dive
→ Crossplane composition deep dive







Configuration Management → Ansible

⑩ OBservability

↳ Monitoring → Nagios
→ Prometheus → TSDB
→ Thanos for HA → Grafana visualization

Logging → Loki
→ Elastic







Tracing → Jaeger




Profiling → Parca

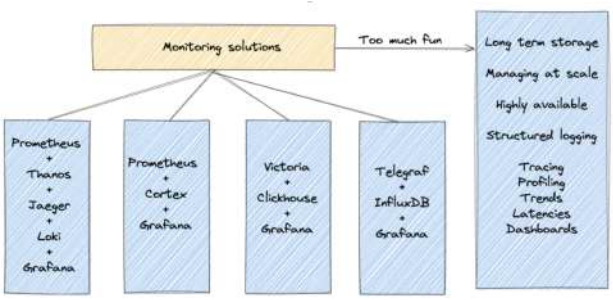
- What is Observability?
- Observability Pillars
- Tools
  ↳ setup, configuration
- OpenTelemetry
- Opentracing

Resources








(11) Chaos Engineering

↳ chaos mesh

Litmus













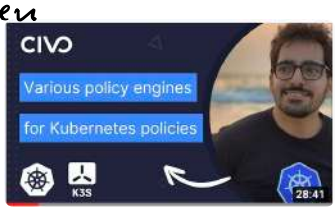
(12) Policy

↳ Kubewarden

↳ OPA

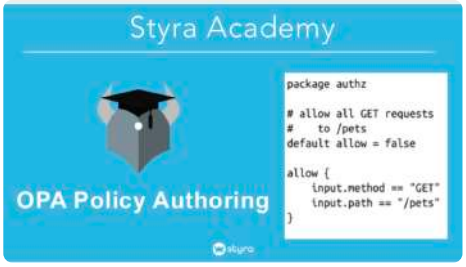
↳ Kyverno

↳ js Policy










(13) Service Mesh



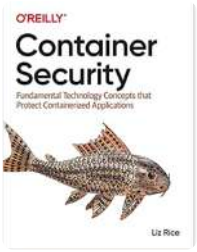

Istio

Linkerd

(14) Others → Supply chain Security → CNCF Landscape

Kubernetes Security

↳ CKS




→ Operators

↳ what are they

how to write Kubernetes operator

↳ Operator SDK

KE

VI

YOU CAN DO IT!!!

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SHARE THIS ROADMAP

GET INTO THE WORLD OF TECH